

ZIYI ZHU

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EXPERIENCE

Slingshot AI

Member of Technical Staff

New York, United States

Nov. 2024 – Present

- Led end-to-end development of a psychology foundation model, spanning **pre-training** on human therapy transcripts, **super-vised fine-tuning** on golden conversations curated by clinical teams, and **alignment** using annotator-labelled preference datasets with DPO, PPO, and GRPO.
- Built the training stack from data preparation and cleaning through multi-GPU distributed training, using TRL and verl for RL; scaled SFT and DPO to **16k context** on **122B-parameter** models with PEFT.
- Deployed language models and reward models (token- and sequence-classification) with vLLM and TensorRT, serving both production traffic and development workloads.
- Trained value models to predict user engagement and retention metrics, applying offline RL against the learned value functions for a **25% increase in user engagement** (words spoken per session).
- Introduced training methods, including synthetic preference data selected by learned value functions and adversarial training for realistic user simulation (DIAL), enabling offline evaluation and multi-turn policy optimisation.
- Led the design and development of an agentic prompt optimisation system (SAGE) in which agents perform diagnostic code execution and propose targeted prompt changes, driving a **+13.7% D1 retention improvement** over the quarter.
- Designed CyclicJudge, a round-robin assignment of LLM judges to evaluation scenarios that recovers panel-mean scores at single-judge cost, mitigating judge bias in model evaluation.
- Improved voice turn-taking by training a Whisper-based audio endpointing model to detect end of turn, **reducing interruption rates by 70%** at equivalent latency.
- Built internal tooling for annotation workflows and benchmark creation using TypeScript and Kotlin.

Unlikely AI

Senior Applied Scientist

London, United Kingdom

Jan. 2023 – Nov. 2024

- Architected core components of a neurosymbolic AI platform combining LLMs with symbolic reasoning, prioritising accuracy, trustworthiness, and explainability for high-stakes decision-making in regulated industries.
- Built semantic parsing and entity resolution models, achieving a **50% improvement** over open-source alternatives through transformer fine-tuning on synthetic datasets for insurance and financial services applications.
- Redesigned the symbolic language layer by integrating LLMs with symbolic solvers, delivering **99% precision** on tasks with **90% cost reduction**; this technology formed the foundation of the platform sold to Lloyds Banking Group.
- Reduced reasoning complexity by optimising ontological reasoning with imitation and reinforcement learning, enabling real-time inference at scale.

Chai Research

Founding AI Researcher

Palo Alto, United States

Dec. 2021 – Nov. 2022

- Built a social AI platform from inception to **200k+ daily active users** and **1M+ daily messages**, one of the largest real-world LLM deployments at the time.
- Drove **60% D1 retention** through LLM fine-tuning and prompt engineering on curated conversational datasets, along with user-centric research and implementation of **20+ app features** in Flutter.
- Trained encoder-based reward models trained on human preference data with rejection sampling, **doubling user engagement** as measured by median conversation length.
- Developed and deployed an offline reinforcement learning pipeline using PPO to maximise user engagement, an early application of RLHF to social AI.
- Deployed LLMs at scale on CoreWeave GPU clusters, building inference infrastructure serving hundreds of models across thousands of GPUs.

Seamless Capital

Quantitative Researcher

Cambridge, United Kingdom

Aug. 2021 – Jan. 2022

- Developed machine learning models (GBDT, MLP, transformers, and ensemble methods) for multi-market horse racing prediction, including Win/Place/Quinella, deployed in live automated trading on Betfair exchange.
- Engineered **50+ alpha signals** and specialised features through historical data analysis, time-series feature extraction, and odds-implied probability calibration, resulting in **10% P&L improvement**.
- Built back-testing infrastructure with cross-validation, walk-forward optimisation, and Monte Carlo simulation for risk analysis, validated through Sharpe ratio, drawdown metrics, and live market execution performance.

UBS Group AG

Financing Quant Intern

London, United Kingdom

June 2021 – Aug. 2021

- Optimised Stock Borrow Loan (SBL) desk operations through automated trade booking, position reconciliation, and settlement.

- Developed analytical tools on the Advanced Cloud Quantitative Analytics (ACQA) platform for repo rate curve construction, collateral valuation, and securities pricing models.

PUBLICATIONS

- **Z. Zhu** et al. “Optimizing Sparse Outcomes Through Dense Behavioral Signals via Value-Guided Preference Distillation.” *Preprint*, 2026. [[arXiv:2609.14648](#)]
- **Z. Zhu** et al. “CyclicJudge: Mitigating Judge Bias Efficiently in LLM-based Evaluation.” *EMNLP 2026 (Main Conference)*. [[arXiv:2603.01865](#)]
- **Z. Zhu** et al. “DIAL: Direct Iterative Adversarial Learning for Realistic Multi-Turn Dialogue Simulation.” *Findings of EMNLP 2026*. [[arXiv:2512.20773](#)]
- **Z. Zhu** et al. “SAGE: Stochastic Prompt Optimization via Agent-Guided Exploration.” *EMNLP 2026 (Industry Track)*. [[arXiv:2606.18902](#)]
- R. Irvine, D. Boubert, V. Raina, A. Liusie, **Z. Zhu**, et al. “Rewarding Chatbots for Real-World Engagement with Millions of Users.” *Preprint*, 2023. [[arXiv:2303.06135](#)]

EDUCATION

University of Cambridge, Christ’s College

Cambridge, United Kingdom

Master of Arts (Cantab.)

May 2025

Bachelor of Arts, Engineering

Oct. 2018 – June 2021

- Results: First Class Honours, ranking in the **top 4%** (**12th** out of **288** students), awarded the Christ’s College Scholarship for **three consecutive years**.
- Modules: Information and Computer Engineering, Electrical and Information Sciences, and Instrumentation and Control.
- Awards: Silver Awards, Singapore Physics Olympiad (2014, 2016); CUES Hackathon Champion 2021; Christ’s College Art Prize 2021; VOMG Most Promising Musician 2017.

COMPETITIONS & PROJECTS

Smartathon (Smart City Challenge) by Saudi Data & AI Authority (SDAIA)

Riyadh, Saudi Arabia

Second Prize Winner

Apr. 2023

- Secured **2nd place** among **14,731** teams globally, earning SAR **97,500** in prize money and presenting the solution at Riyadh.
- Built a pothole severity classification system combining YOLOv5 for real-time 2D object detection and DiffNet for monocular depth estimation and 3D surface reconstruction, enabling automated road maintenance optimisation.

Personal Projects

- Engineered and deployed a profitable tennis match prediction system on Betfair, consistently achieving **~5%** daily returns.
- Derived closed-form probability model for stop-limit order execution using Optional Stopping Theorem and exponential martingales on geometric Brownian motion, validated through Monte Carlo simulations, achieving **<1%** relative error.

SKILLS & INTERESTS

LLM Training: PyTorch, Hugging Face Transformers, TRL, verl, DeepSpeed; post-training (SFT, DPO, PPO, GRPO); PEFT (LoRA, QLoRA)

LLM Inference: vLLM, TensorRT

Programming: Python, TypeScript, JavaScript, Kotlin, Java, C++, SQL, Flutter, React, MATLAB

Domains: Natural Language Processing, Computer Vision, Diffusion Models, Audio Processing (speech endpointing & generation)

Classical ML & Vision: XGBoost, LightGBM, OpenCV, scikit-learn, image segmentation, depth estimation

Infrastructure: Distributed training (multi-node, multi-GPU), Docker, Kubernetes, GCP, AWS, Weights & Biases, Nebius Cloud

Languages: Chinese (Native), English (Fluent), Japanese (Beginner)

Interests: Percussive fingerstyle guitar, piano, drum set, visual art, oil painting, photography, pool, tennis, table tennis, and travel